

NGO Connect App – Complete Project Documentation

This document combines all essential information about the NGO Connect app, including features, technologies, and extension guidance. Use this for your project submission and viva preparation.

1. App Overview

NGO Connect is a Flutter-based platform connecting NGOs, volunteers, and administrators. It streamlines campaign management, volunteer coordination, donations, analytics, and community engagement, maximizing social impact.

User Types:

- **NGO**: Manage campaigns, volunteers, donations, and public profile.
- **Volunteer**: Discover NGOs, join campaigns, donate, track impact, and participate in events.
- **Admin**: Oversee NGO registrations, approve/reject applications, and maintain platform integrity.

2. Features by User Type

A. NGO Side

- Dashboard, campaign management, volunteer management, donation tracking, SOS alerts, quick tasks, analytics, marketplace, notifications, public profile.

B. Volunteer Side

- Discover NGOs, join campaigns, donate, find opportunities, community, emergency response, impact tracking, event calendar, government schemes, ask experts.

C. Admin Side

- Dashboard for NGO registration review, approval/rejection, statistics, and document verification.

3. Shared Features

- Authentication, location services, media support, push notifications, community posts, CSR integration, needs forecasting.

4. How Major Functions Work

- Authentication: Firebase Auth for secure login and session management.

- Campaign Management: Firestore for campaign data, Flutter UI for creation/joining.

- Volunteer Management: Firestore and Cloud Functions for applications, approvals, and notifications.

- Donation Tracking: Razorpay for payments, Firestore for records.

- SOS Alerts: Firestore and FCM for real-time emergency notifications.

- Analytics: Data aggregation and visualization with Firestore and fl_chart.

- Marketplace: Firestore for resource sharing.

- Notifications: FCM for real-time updates.

- Public Profile: Firestore and Flutter UI for NGO showcase.

5. Technologies and Integration

- Flutter: Cross-platform UI.

- Firebase: Auth, Firestore, Storage, Cloud Functions, Messaging.

- Razorpay: Payment gateway.
- Google Maps: Location services.
- flchart, imagepicker, audioplayers, vibration, etc.

Integration:

- UI in Flutter, data in Firestore, files in Storage, notifications via FCM, payments via Razorpay, maps via Google Maps.

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<h2>6. How to Add or Extend Functionalities</h2>

Plan the feature, update Firestore, create/update UI, add/update services, integrate with APIs, test, and document.

NGO: Add campaign types, analytics, resource sharing.

Volunteer: Add skill-based volunteering, new engagement modules, impact tracking.

Admin: Add verification steps, analytics, automated workflows.

Follow modular structure, reuse widgets, centralize logic, secure data, and document changes.

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<h2>7. Security & Best Practices</h2>

All sensitive data is secured via Firebase rules.

User authentication is required for all actions.

Data validation is enforced both client-side and server-side.

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<h2>8. Project Structure (lib/)</h2>

main.dart: App entry point

firebase_options.dart: Firebase config

screens/: UI screens for admin, ngo, volunteer

services/: Business logic (auth, notifications, cache)

utils/: Helper utilities

widgets/: Reusable UI components

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<h2>9. Contribution & Extension</h2>

Follow modular structure for new features.

Use services for business logic.

Write reusable widgets for UI consistency.

Update documentation for any new modules.

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For more details, see the individual documentation files in the project root.

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Prepared for project submission and viva. All the best!